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XY Gantry SYSTEM

MECHATRONICS SYSTEM DESIGN

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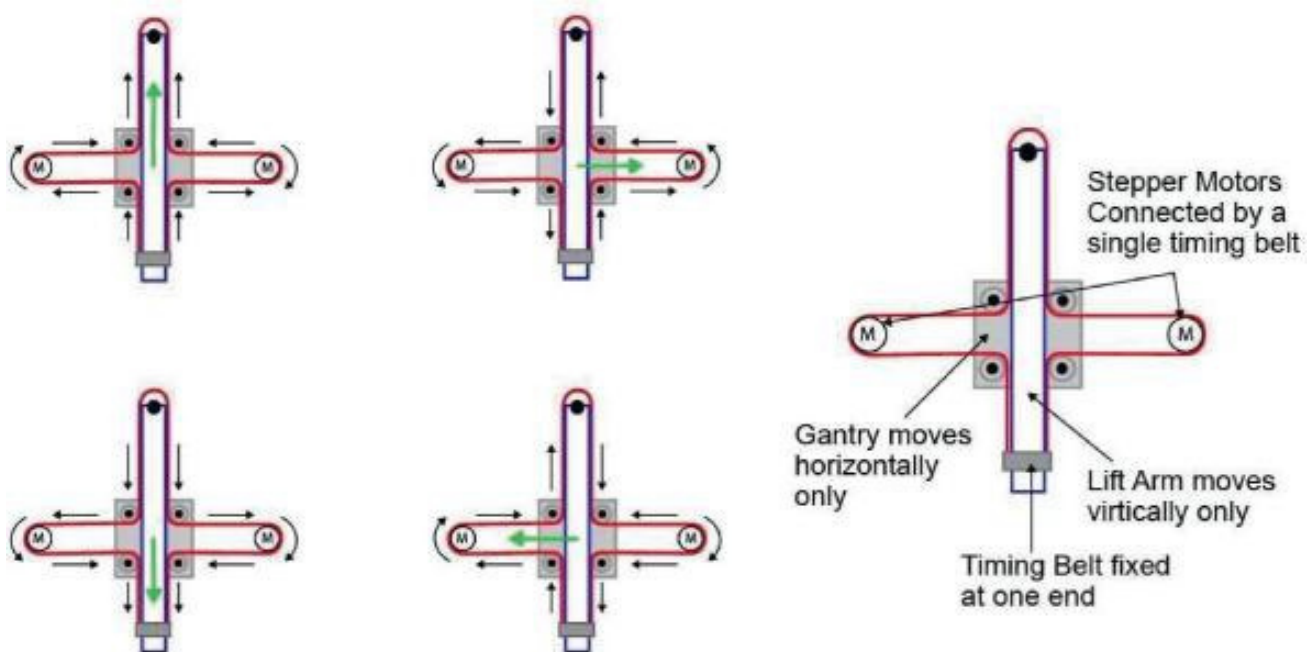
Dr. Abdelhalim bassiouny

Project's Aim:

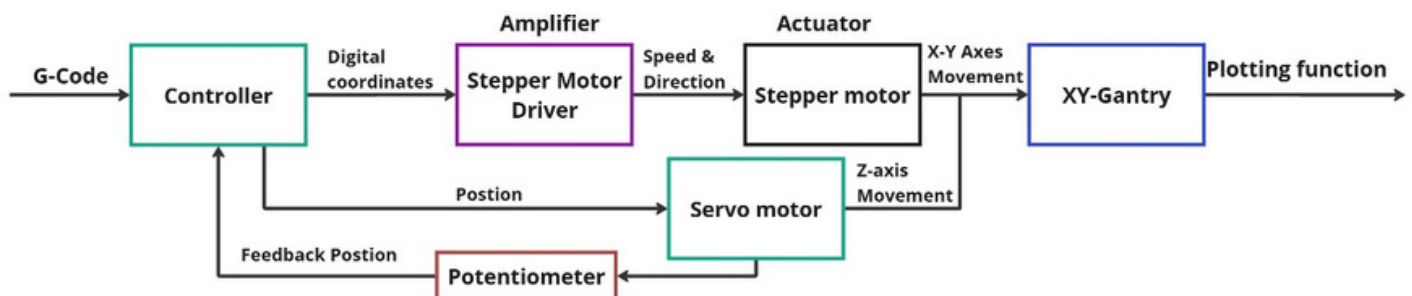
This project aims to create a computer numerically controlled (CNC) plotter designed to produce accurate drawings based on 2D CAD files. The system translates CAD designs into digital coordinate commands, which are executed by the end-effector (pen) through a mechanism powered by stepper motors and timing belts, ensuring accurate replicate of the 2D designs.

Working Principle:

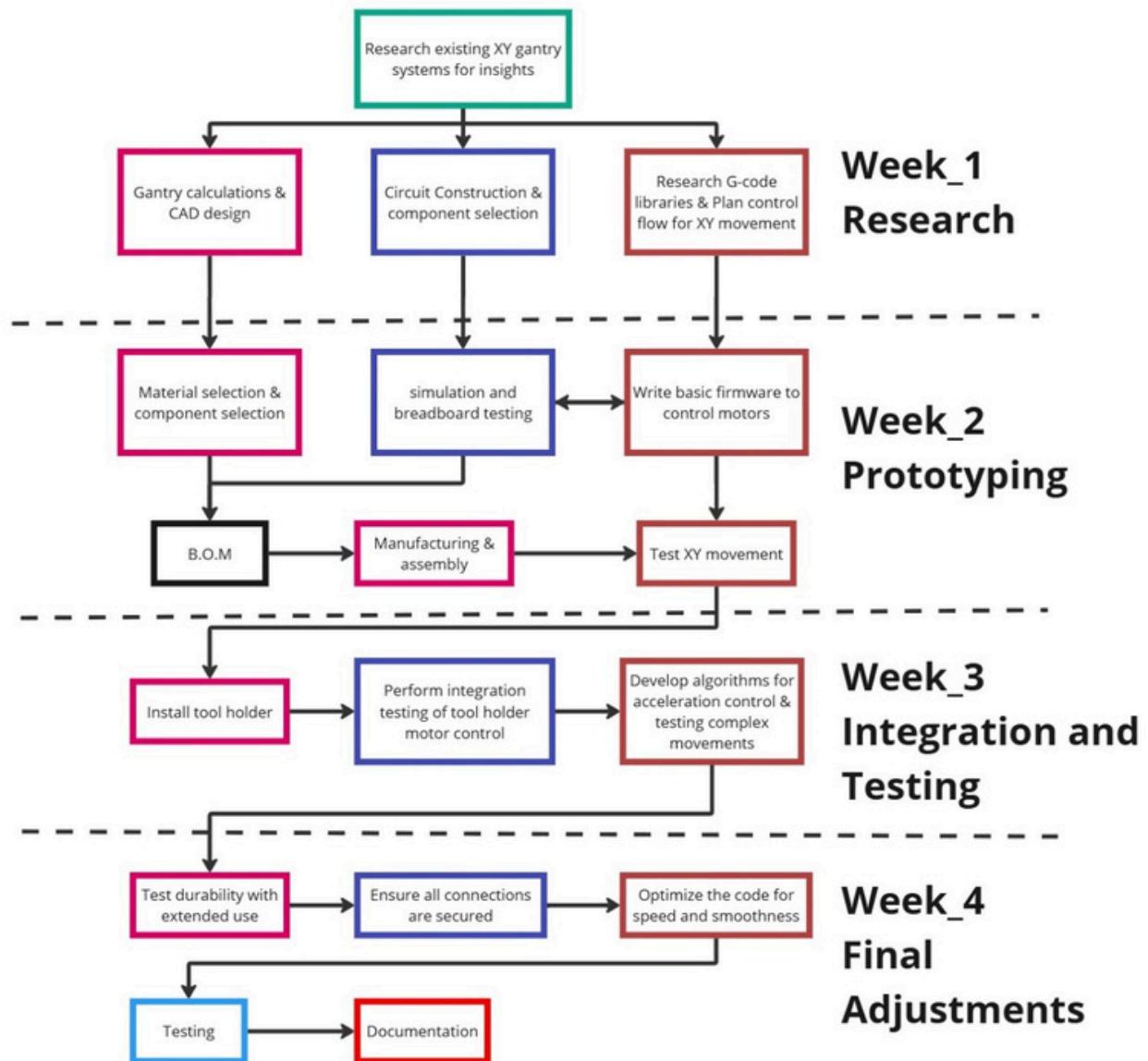
The xy gantry system depends on a single **timing belt** that is connected in a continuous loop and mounted on two **stepper motors** to control the X-axis movement and the Y-axis movement based on the direction of rotation of the stepper motors, if the motors rotate in the same direction the gantry will move along X-axis, if the motors rotate in the opposite direction the gantry will move along the Y-axis.



Mechatronics XY-Gantry Block Diagram:

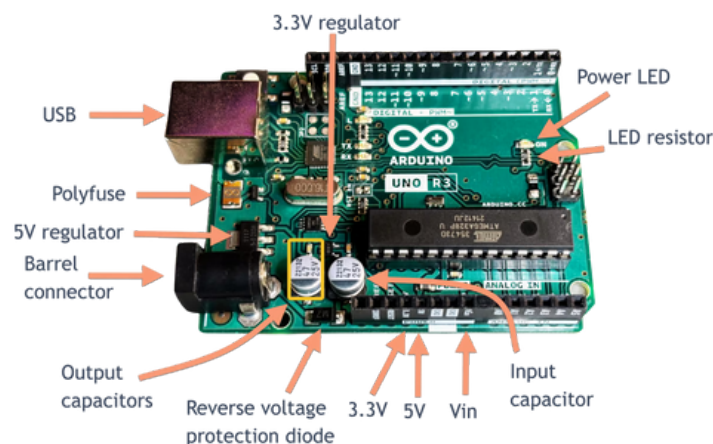


Procces Flow Diagram:



Electrical Components:

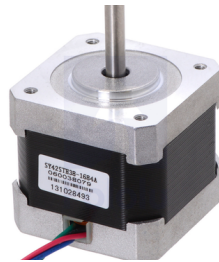
- **Arduino (uno)** : processing the 2D design and sending signals to the stepper motor drivers to control the stepper motors movement along the X and Y axes.



- **DRV8825 stepper motor driver:** convert pulse signals from the controller into motor motion to achieve precise positioning
- **Nema 17 stepper motor:** actuate the XY- gantry through rotational motion
- **Servo motor (SG90):** Responsible for the end-effector movement in the Z-axis
- **Limit switches:** Define limits in order to prevent a CNC from damaging itself if it travels past its limits
- **CNC expansion shield:** simplify and facilitate the connection of stepper motors, drivers, and control signals to a microcontroller
- **Power supply:** provide sufficient power for the system



DRV8825



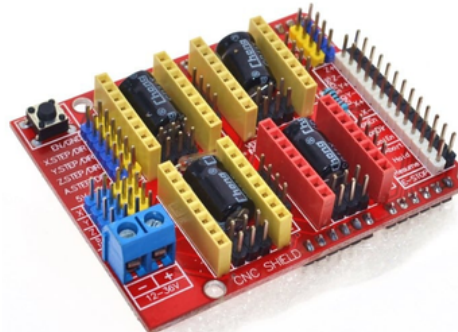
Nema17 Stepper motor



SG90 Servo Motor

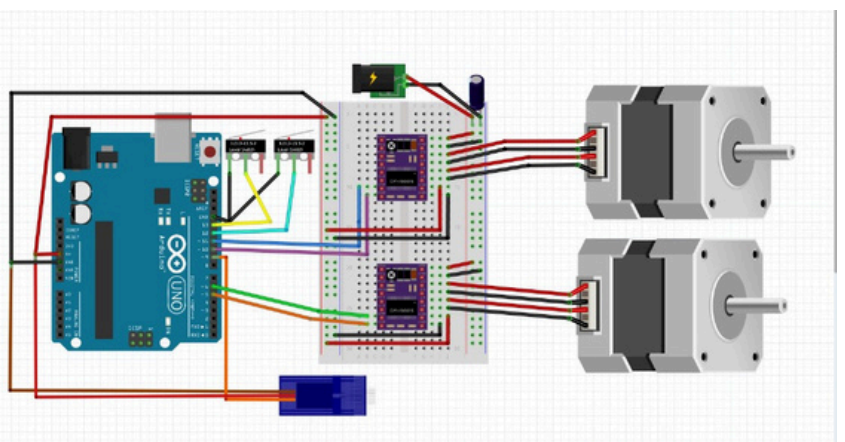
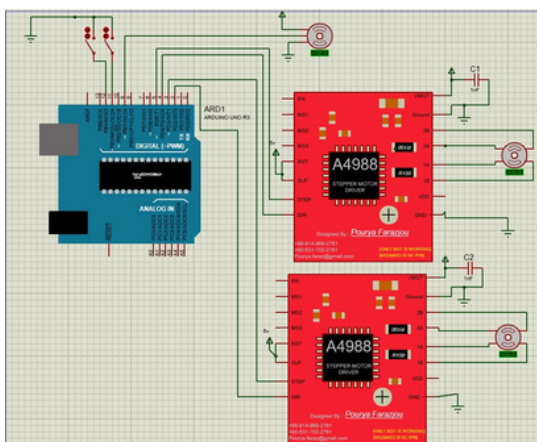


Limit Switches



CNC Expansion Board

Circuit Diagram:



Power Transsmision elements:

The difference between the belt and lead screw for linear motion and which one of them is suitable for XY Gantry.

	Belts	Lead Screws
Speed & Acceleration	High speed & acceleration. Ideal for application requiring quick movement over longer distance.	Slower compared to belts. Better suited for precise and controlled movement rather than speed.
Load capacity	Low load capacity	High load capacity
Precision & Accuracy	Less precise due to potential belt stretch and backlash	High precision, especially with anti-backlash nuts.
Efficiency	Highly efficient, with minimal energy loss. And lower friction than lead screws.	Less efficient, as friction between the screw and nut consumes energy. It requires lubrication for smoother operation.
Durability and Wear	Subject to wear, especially if over-tensioned or exposed to debris.	Durable, but wear can occur on the nut or screw threads over time.
Maintenance	It is simple and low-maintenance. And easy to replace or adjust tension.	It requires regular lubrication and inspection for wear. The replacement can be more complex.
Nois	Quiet operation due to smooth motion.	Can be noisier, especially if poorly lubricated or under heavy loads.
Cost	Lower cost initially. As it is economical for long-distance or lightweight applications.	Higher cost, especially for high-precision screws.
Application	Any system needing fast, smooth motion over longer distances like 3d printer,	Application requiring high precision and load capacity

According to this comparison, the belt is suitable mechanism for XY Gantry applications.

Advantages of Timing Belts for XY Plotters:

- **High Precision:** The toothed design ensures there is no slippage between the belt and pulley, allowing for precise motion, which is critical in applications like plotting, cutting, or 3D printing.
- **Smooth Motion:** The continuous engagement of teeth with pulleys results in smooth and consistent movement, which is essential for maintaining drawing or cutting quality.
- **High-Speed Capability:** Timing belts are lightweight and can move quickly, making them suitable for rapid positioning without losing accuracy.
- **Durability:** Made from materials like polyurethane or reinforced rubber, they resist wear and provide a long service life under normal operating conditions.
- **Silent Operation:** Compared to other mechanisms like chains or gears, timing belts are quiet, an advantage in environments where noise reduction is desirable.

Mechanical Components:

- **Timing Belts:** Feature teeth that mesh with pulleys to prevent slippage, ensuring precise positioning.
- **Pulleys:** Fixed to the shafts of stepper motors or idler shafts
- **Shaft Rods:** Provide a path for the base of the xy gantry to move along
- **Linear Bearing:** provide free motion in both directions along the shaft rods



GT2 Timing Belt



GT2 Pulley



LM8uu Linear Bearing



Shaft Rods

Motion requirement

Stroke (X axis) = 420 mm – 78mm = 342 mm

Stroke (Y axis) = 402.5 mm - 78 mm = 324.5 mm

Speed = 0.3 m/sec

Acceleration = 0.45 m²/sec

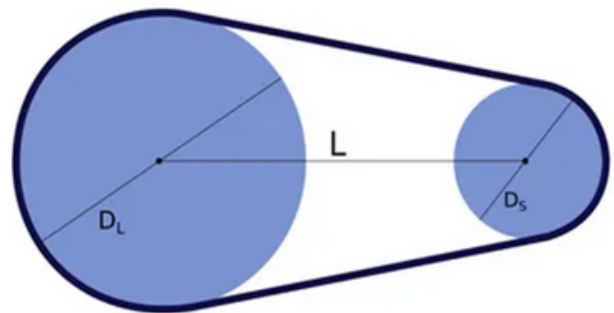
Load = 0.5 Kg

Belt Calculations

Belts that connect two pulleys together must have their specified tension to maintain the necessary friction between the belt and the pulleys to transmit torque successfully. We can use belts to link two pulleys of the same or different sizes, and we always do so on pulleys of the same surface.

Using longer belts than needed could result in the belts getting detached from the assembly. On the other hand, excessively forcing a short belt to fit a pulley system can lead to premature breakage of the belt. That is why we need to know precisely the correct belt length, just like getting the right gear sizes for a gear train.

$$L_{\text{belt}} = \frac{\pi \times (D_L + D_S)}{2} + (D_L - D_S) \times \arcsin \left(\frac{D_L - D_S}{2 \times L} \right) + 2 \times \sqrt{L^2 - \frac{(D_L - D_S)^2}{4}}$$



$D_L = D_H = 12 \text{ mm}$

$L_x = 486 \text{ mm}$

$L_y = 405.5 \text{ mm}$

- belt length in x-axis $L_{\text{belt}} = 1009.7 \text{ mm}$
- belt length in y-axis $L_{\text{belt}} = 848.7 \text{ mm}$

But due to the working principle that we use then the belt length should be used is 1791.77 mm. (this dimension is taken from SolidWorks after finishing the design)

Belt Tension

Proper tension is required to avoid slipping or sagging:

$$T = F + \frac{m \cdot a}{2}$$

$$T = m \cdot g \cdot \mu + \frac{1}{2} m \cdot a = 2.6 \text{ N}$$

where:

- T = tension in the belt.
- F = force required to move the load.
- m = total mass of the moving parts.
- a = acceleration.

Belt thickness based on tension capacity

The belt thickness t can be approximated for flat belts based on tensile stress:

$$t = \sqrt{\frac{T}{b \cdot \sigma_t}}$$

$$t = 0.011 \text{ m} = 11 \text{ mm}$$

Where:

- T: Maximum tension in the belt (N).
- b: Belt width (m). = 6mm
- σ_t : Allowable tensile stress of the belt material (Pa). = 3448378.65 Pa

Motion Profile

Angular Velocity:

$$W_{\max} = V / r_{\text{pulley}} = 0.3 / 0.006 = 50 \text{ rad/sec}$$

Angular Acceleration:

$$\alpha_{\max} = w_{\max} / t = 50 / 0.2 = 250 \text{ rad/sec}^2$$

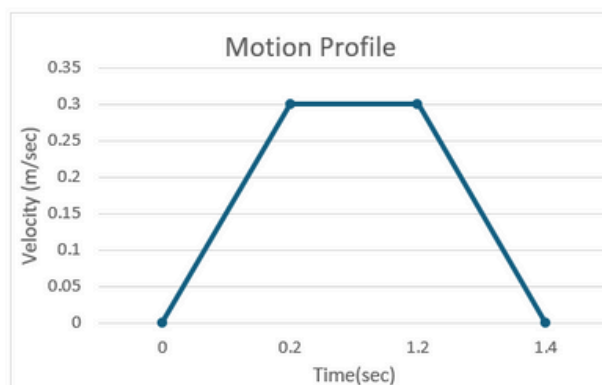
Inertia

Motor & Load side pulley inertia:

$$J_{pm} = J_l = \frac{1}{2} m_{\text{pulley}} \cdot r_{\text{pulley}}^2 = \frac{1}{2} \cdot 0.0054 \cdot 0.006^2 = 9.72 \times 10^{-8} \text{ Kg.m}^2$$

Load side pullet inertia reflected to Motor:

$$J_{pl \rightarrow m} = (J_{pl} / e)(1/N_r)^2 = (9.72 \times 10^{-8} / 0.9) (1/1)^2 = 1.08 \times 10^{-7} \text{ Kg.m}^2$$



Belt inertia reflected to Motor:

$$J_{B-m} = (m_B * r^2) / e = (0.017 * 0.006^2) / 0.5 = 6.8 \times 10^{-7} \text{ Kg.m}^2$$

Load inertia reflected to Motor:

$$J_{L-m} = (m_L * r^2) / e = (0.5 * 0.006^2) / 0.9 = 2 \times 10^{-5} \text{ Kg.m}^2$$

Total inertia:

$$J_{\text{total}} = J_{pm} + J_{pl-m} + J_{B-m} + J_{L-m} = 2.09 \times 10^{-5} \text{ Kg.m}^2$$

Torque required

Torque due to constant motion: $T_c = T_{\text{friction}} = m_{\text{load}} * g * \mu * r_{\text{pulley}} = 0.5 * 9.81 * 0.5 * 0.006 = 0.015 \text{ Nm}$

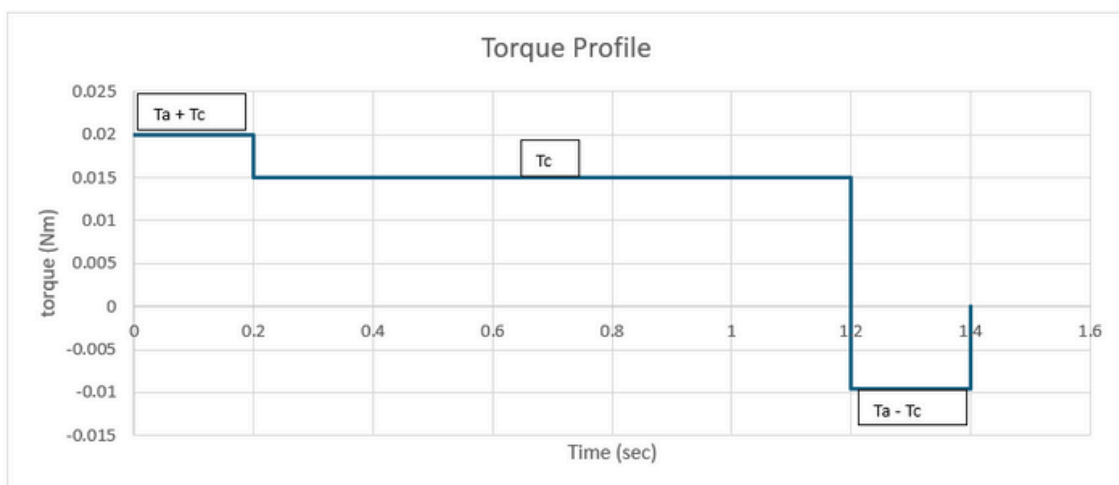
μ : friction coefficient (0.3 – 0.5)

Torque due to acceleration: $T_a = \alpha_{\text{max}} * J_{\text{total}} = 250 * 2.09 \times 10^{-5} = 5.225 \times 10^{-3} \text{ Nm}$

$$T_{\text{max}} = T_c + T_a = 0.015 + 5.225 \times 10^{-3} = 0.2 \text{ Nm}$$

Torque rms:

$$T_{\text{rms}} = \sqrt{\frac{T_1^2 t_1 + T_2^2 t_2 + T_3^2 t_3 + T_4^2 t_4}{t_1 + t_2 + t_3 + t_4}} \quad T_{\text{rms}} = 0.15 \text{ Nm}$$



then the required torque of motor is 0.2 Nm

Step Resolution

To calculate the linear movement per step of the stepper motor:

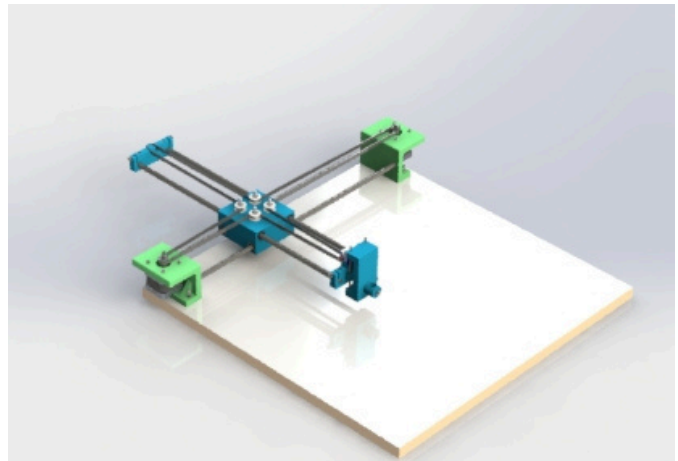
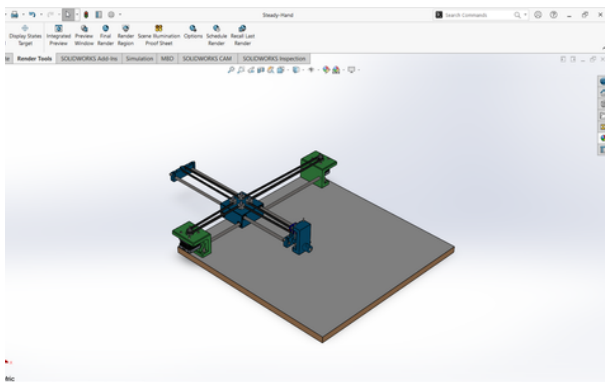
$$\text{Linear Distance Per Step} = \frac{\pi \cdot D}{\text{Steps Per Revolution}}$$

Where:

- D: Diameter of the pulley
- Steps Per Revolution: Number of steps in one motor revolution (e.g., 200 for a 1.8° stepper motor).

If micro stepping is used, adjust the resolution accordingly.

Concept Design:

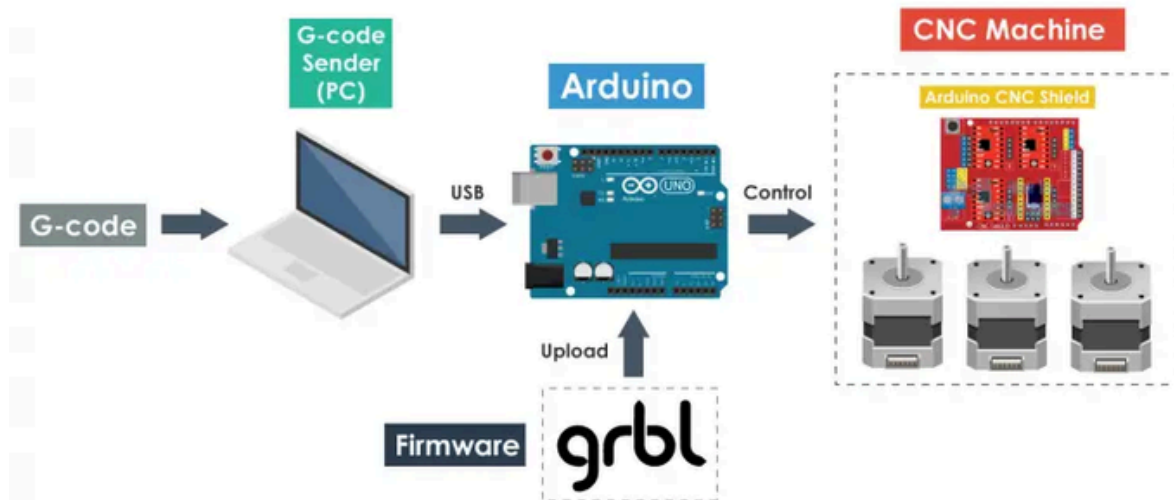


B.O.M Sheet:

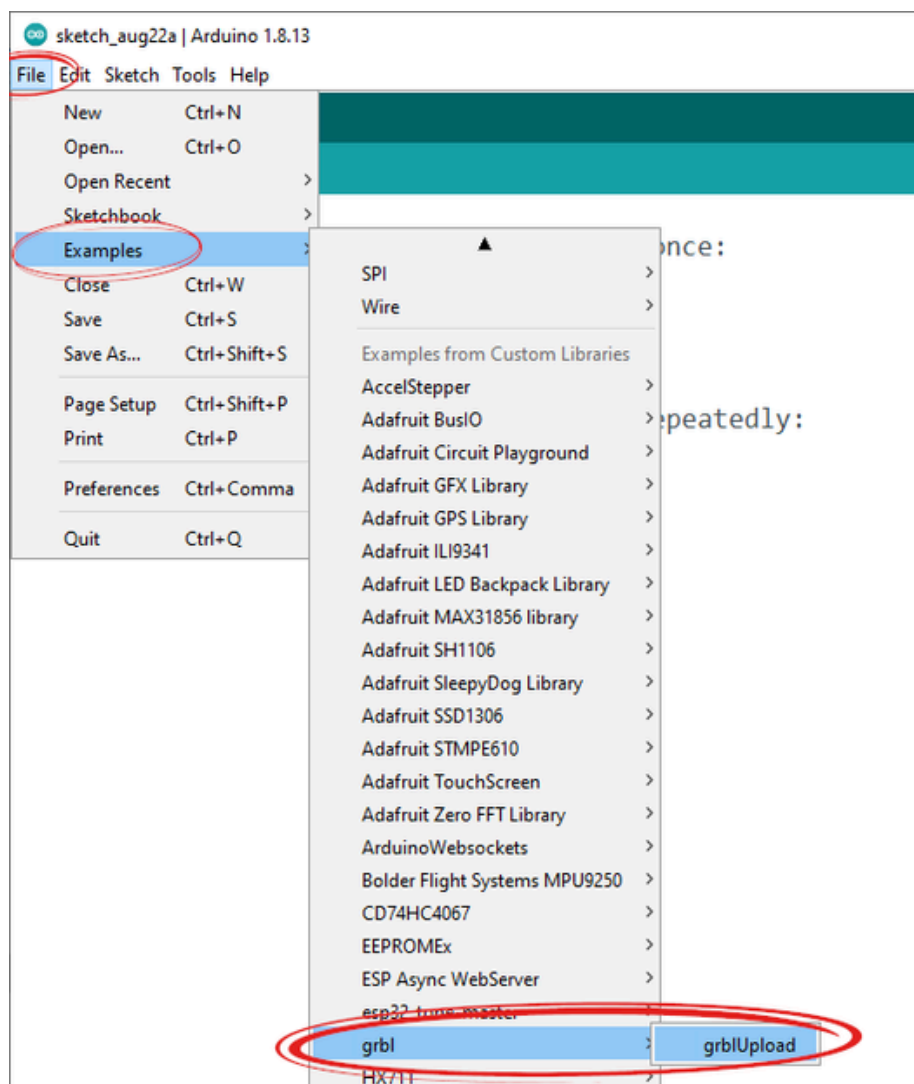
XY gantry B.O.M					
File Edit View Insert Format Data Tools Extensions Help					
100% \$ % .0 .00 123 Defaul... - 10 + B I ÷					
A1	Component				
	A	B	C	D	E
1	Component	Quantity	price per part (EGP)	total price (EGP)	Vendor
2	Arduino Uno	1	430	430	Makers Electronics
3	Nema 17 Stepper Motor	2	550	1100	Makers Electronics
4	Servo motor (SG90)	1	90	90	Makers Electronics
5	DRV8825 Stepper Motor Driver	2	115	230	Makers Electronics
6	Limit switch	2	8	16	Makers Electronics
7	CNC Expasion shield	1	115	115	Makers Electronics
8	Linear Rod M8 x 450mm	2	150	300	Most Electronics
9	Linear Motion Bearing LM8UU / 8mm	4	40	160	Most Electronics
10	GT2 Pulley	2	40	80	Most Electronics
11	GT2 Timing belt	2	50	100	Most Electronics
12	Screws & Fastners	35	1.5	52.5	Most Electronics
13	3D printing fabrication	190 (gram)	3	570	Metanoia
14					
15				Total Cost	3243.5

Software:

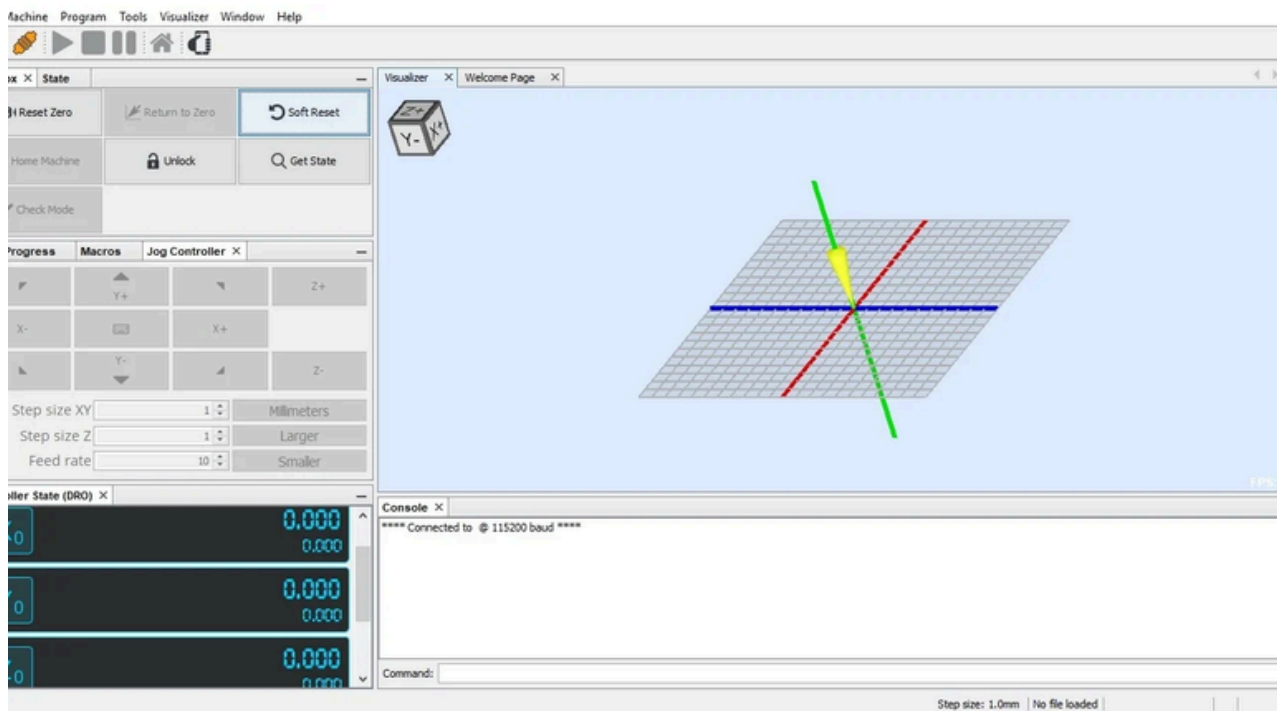
In order to translate the G-code into Motor movements we need to install **GRBL** (G-code reference block library) firmware on the Arduino development board.



As shown in the diagram above the whole purpose of the GRBL firmware is to give the arduino the ability to translate the G-code (Input) into motor movements (Output). The GRBL firmware could be easily downloaded and and uploaded to the arduino through the arduino IDE.



After giving the arduino the ability to translate G-code, a software is needed to send a g-code to the arduino in order to move the stepper motors accordingly. The software required is called **UGS** (Universal G-code sender) which is also considered as a GRBL controller



UGS software can also be used to configure the GRBL parameters, adjust the steps/mm parameter, configure the limit switch states and adjusting the auto-home position.

Now all what is needed is to generate a g-code to be passed on the UGS software in order to be translated into motor movements. The software used to generate g-code from 2D files is called **Inkscape**

